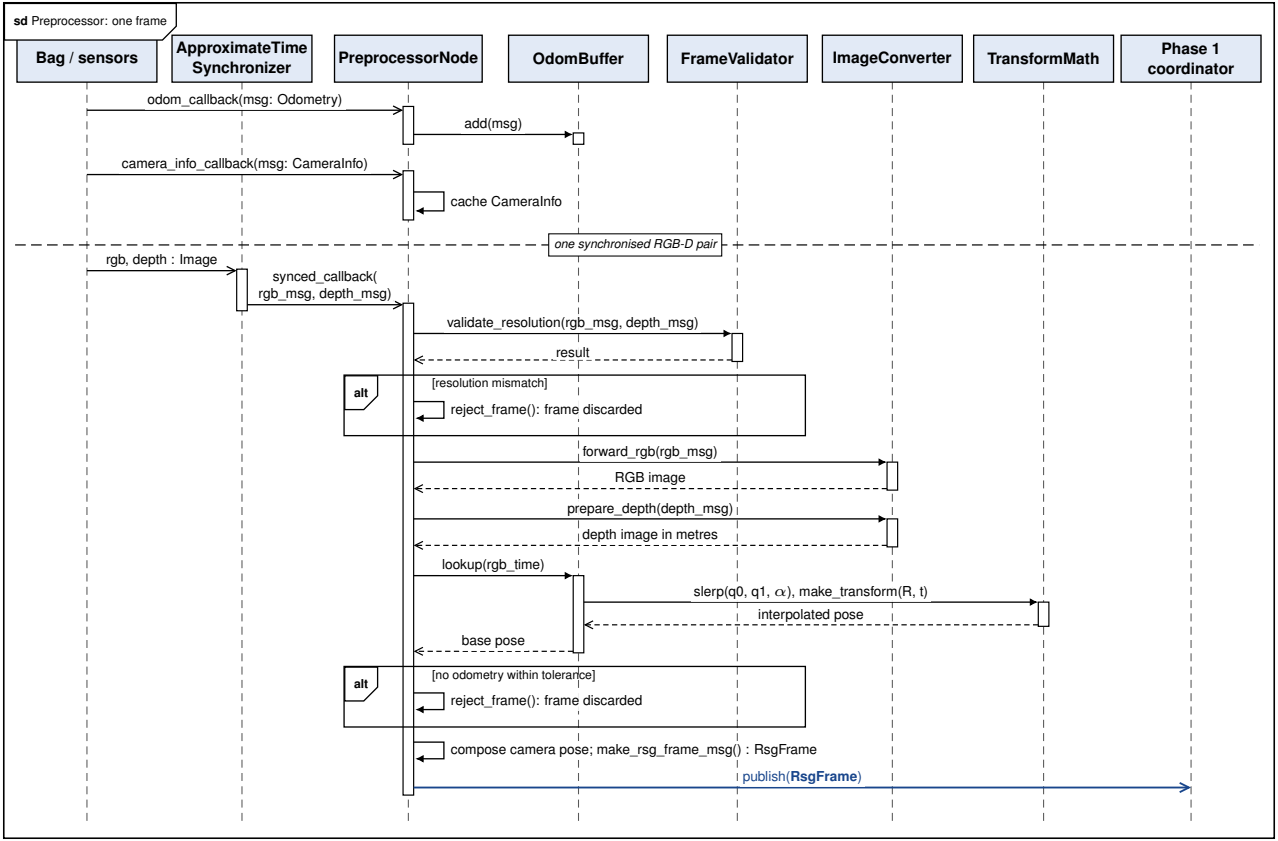
sd Preprocessor: one frame
Bag / sensors
ApproximateTime Synchronizer
PreprocessorNode
OdomBuffer
FrameValidator
ImageConverter
TransformMath
Phase 1 coordinator
odom_callback(msg: Odometry)
add(msg)
camera_info_callback(msg: CameraInfo)
cache CameraInfo
one synchronised RGB-D pair
rgb, depth : Image
synced_callback(rgb_msg, depth_msg)
validate_resolution(rgb_msg, depth_msg)
result
alt
[resolution mismatch]
reject_frame(): frame discarded
forward_rgb(rgb_msg)
RGB image
prepare_depth(depth_msg)
depth image in metres
lookup(rgb_time)
slerp(q0, q1, α), make_transform(R, t)
interpolated pose
base pose
alt
[no odometry within tolerance]
reject_frame(): frame discarded
compose camera pose; make_rsg_frame_msg() : RsgFrame
publish(RsgFrame)